OM BHARADWAJ

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SUMMARY

Computer Science undergraduate at IIIT Manipur (CGPA 9.05) who builds medical imaging models and turns them into working apps. Ocula, a diabetic retinopathy screening platform, is the basis of my first-author paper accepted at IEEE SILCON. Cutis screens skin lesions on ISIC 2024 and adds explainable predictions and RAG reports grounded in clinical guidelines. At Tata Technologies I worked on an enterprise CRM portal. Looking for software engineering, backend or applied AI roles.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, Java, SQL, PHP, C
Backend & Web: FastAPI, Node.js, Express.js, PHP, REST APIs, Microservices, JWT auth, role-based access control, React.js
Databases: PostgreSQL, MySQL, MongoDB, SQLite
Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, Random Forest, SVM, ADASYN, Optuna, Pandas, NumPy
Models: EfficientNet-B4, ResNet-18, DenseNet-121, GoogLeNet, ConvNeXt, Swin-T, EdgeNeXt, EVA-02
Explainability, LLMs & RAG: Grad-CAM, Score-CAM, LIME, SHAP, Groq LLaMA-3, Gemini API, hybrid BM25 + embedding search
DevOps & Testing: Docker, Docker Compose, Nginx, Git, GitHub, CI/CD, Render, automated testing
Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

EXPERIENCE

Tata Technologies May 2026 to July 2026
Software Engineering Intern *Jamshedpur, Jharkhand, India*

- Worked in an agile team on an enterprise CRM portal that large clients use to manage their customer relationships.
- Built a responsive React.js frontend with dashboards, charts and real-time notifications.
- Wrote REST APIs in PHP with role-based access control, authentication middleware and automated reporting.
- Designed MySQL schemas with normalized tables, indexed queries and stored procedures so the system could scale.

PROJECTS

Ocula: Diabetic Retinopathy Screening Platform Demo Video | GitHub
Python, PyTorch, FastAPI, Node.js/Express, React, PostgreSQL, Docker

- Built a full-stack diabetic retinopathy screening platform on a validated clinical dataset of **3,662 images**, deployed as four Docker Compose services with JWT auth and role-based access.
- Reached **98.50% binary accuracy** (AUC 0.997) and **90.68% five-class accuracy** on APTOS 2019, **0.70 and 1.39 points above** the Butt et al. (2022) baseline.
- The work is the basis of my first-author paper, **accepted at IEEE SILCON**.
- Ensembled five CNN backbones with 8-view test-time augmentation and soft-voting SVM, RF and XGBoost, lifting binary accuracy **1.38 points** over the best single backbone.
- Designed a two-level hierarchical SVM cascade with ADASYN oversampling for five-class grading, a **1.27 point gain** over a flat classifier, and raised Severe DR recall from **0.79 to 0.82**.
- Added Grad-CAM, LIME and SHAP plus Groq LLaMA-3 clinical and patient reports, all in **under 45 seconds on CPU**.

Cutis: AI Skin Lesion Screening Platform Demo Video | GitHub
Python, PyTorch, FastAPI, Groq LLaMA-3, Node.js/Express, React, MySQL, Docker

- Built a full-stack skin lesion screening platform on the **400K+ image** SLICE-3D dataset (**401,059 lesions, 393 malignant**), deployed as four Docker Compose services with JWT auth and patient and clinician roles.
- Reached **0.1683 pAUC80** and **0.9653 AUROC** out-of-fold on the official ISIC 2024 metric (5-fold CV grouped by patient). The ISIC 2024 winner scored 0.1726 and 0.9668 on the hidden test set.
- Ensembled **20 CNNs** (EVA-02-S, EdgeNeXt-B, ConvNeXt-T, Swin-T) and fused them with a tabular GBDT stage and an Optuna-tuned LightGBM head, lifting pAUC80 from **0.1646 to 0.1683** (**0.1647** leave-one-institution-out).
- Flags **32.1 lesions per cancer** found at 80% sensitivity, and ranks **80.8%** of each patient's cancers in their top 15 lesions (winner 0.790).
- Explains each scan with Grad-CAM (Score-CAM fallback), LIME and SHAP. A Groq LLaMA-3 agent writes reports and follow-up answers grounded in NCI PDQ and CDC guidelines (hybrid BM25 + embedding search), citations checked on the server.
- Scores a photo in about **4 seconds on CPU** with all 20 networks in about **2 GB of RAM**, using the image branch only.
- Verified with **106 automated tests** (50 Node, 46 Python, 10 web) against a real MySQL server, plus a replay tool that agrees with stored out-of-fold predictions on 300 lesions (Spearman > 0.8).

EDUCATION

Indian Institute of Information Technology Manipur Imphal, India
B.Tech, Computer Science & Engineering (AI & Data Science) *2023 to 2027 (Expected)*
CGPA: 9.05 / 10.0 (Top 5% of the cohort)

ACHIEVEMENTS & LEADERSHIP

- Top 5% of my cohort at IIIT Manipur across five semesters, with a CGPA of 9.05.
- GDG On Campus Chapter Lead, IIIT Manipur. Selected through Google's competitive process. I run Study Jams, workshops and developer events for 1200+ students.
- GeeksforGeeks Campus Ambassador. I help students find technical resources and get started with competitive programming, and I solve problems regularly on LeetCode.